Tiago Ferrão-Lindqvist

Test & Validation Engineer | Python, SQL & Test Data
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Professional Summary

Mechanical engineer with 12 years of experience in product design and test and validation of automotive and industrial components. Develops Python tools to process test-rig data, check channels against specifications and generate reports, including a nightly workflow handling approximately 2 GB of data per week across six rigs. Experience querying SQL Server, writing results to an IT-created staging table and creating Power BI reporting for engineering users. Currently transitioning toward data engineering and completing an online data engineering course.

Experience

Test & Validation Engineer, Corvane Automotive Components — Setúbal *Mar 2019–Present*

- Plan and run durability, vibration and climatic tests on seat-adjuster mechanisms across six servo-hydraulic and electric test rigs; write reports sent to OEM customers.
- Developed Python scripts using pandas, NumPy, Matplotlib and npTDMS to parse TDMS and CSV rig output, check channels against test specifications and generate standard report figures; reduced report preparation from approximately three days to half a day per test.
- Run the processing workflow nightly on a lab PC using Windows Task Scheduler; it writes summary results to an IT-created staging table in the plant's SQL Server database. Packaged as an internal tool now used by four other test engineers.
- Query the main test-results database in SQL Server for trend analysis across three product generations; the queries are read-only.
- Created a Power BI dashboard using the staging table to show pass/fail trends and cycles to failure by product generation; used by approximately 15 validation and design engineers.
- Correlate rig results with ANSYS Mechanical FEA and lead 8D root-cause analysis on field failures; proposed a redesigned adjuster rail bracket that cut warranty returns for that part by 40%.
- Maintain MATLAB/Simulink models of the adjuster drive used to select rig load profiles.
- Trained three apprentices in rig safety and test setup (2022–2024).

Mechanical Design Engineer, Halvik Pumps — Porto *Sep 2014–Feb 2019*

- Designed centrifugal pump casings and impellers in SolidWorks and prepared production drawings with GD&T; owned DFMEA for two product lines.
- Performed structural FEA of casings in ANSYS; supported a change from grey cast iron to ductile iron for one casing family, saving 12% in material cost.
- Supported conversion of the pump performance test bench to a Siemens S7 PLC-controlled setup, adjusting setpoints and timers in TIA Portal under the automation engineer's supervision.
- Wrote a LabVIEW acquisition program for the performance test bench, reducing test setup time from approximately two hours to 30 minutes.

Selected Project

Bike-sharing trips API | Personal learning project, 2025

- Built a small Flask API with three read-only endpoints over a public bike-sharing dataset loaded into SQLite. Runs on a personal laptop; followed a tutorial to run it with Docker Compose. Used Python, Flask, SQLite, Docker Compose and Git.

Education

MSc Mechanical Engineering, Sado Valley Polytechnic 2014
Thesis: Fatigue life prediction of welded brackets under variable-amplitude loading

Professional Development

Online Data Engineering course — in progress; six of nine modules completed.

Skills

Data processing and reporting: Python, pandas, NumPy, Matplotlib, npTDMS, SQL Server, SQL, Power BI, Excel/VBA
Engineering and testing: MATLAB/Simulink, ANSYS Mechanical, SolidWorks, CATIA V5, LabVIEW, Minitab, design of experiments (DOE), GD&T, DFMEA, 8D problem solving
Personal learning project tools: Flask, SQLite, Docker Compose, Git

Languages

Portuguese (native), English (fluent), Spanish (good)